

Final Report: Reinforcement Learning with Pre-Collected Datasets (RLPD) and Soft Actor-Critic

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Abstract

This report presents experiments using the Soft Actor-Critic (SAC) algorithm combined with Reinforcement Learning from Pre-Collected Datasets (RLPD). We evaluate performance on continuous control environments Hopper-v4 and Ant-v5 and analyze the effects of replay buffer size, batch size, and random seed variation on learning stability and final performance. We additionally evaluate AntMaze, a sparse-reward navigation task, to study the interaction between offline data and stabilization techniques. Overall, incorporating offline data improves early-stage learning stability and reduces variance across seeds, while additional design choices are necessary for reliable performance in long-horizon tasks.

1 Introduction

Reinforcement learning (RL) enables agents to learn optimal behavior through interactions with an environment and reward feedback. However, standard online RL methods are often unstable and data-inefficient, particularly in high-dimensional or sparse-reward settings.

Reinforcement Learning with Pre-Collected Datasets (RLPD) addresses these challenges by seeding the agent’s replay buffer with previously collected trajectories, thereby improving early exploration and reducing variance across training runs.

Prior work has shown that Soft Actor-Critic (SAC) is particularly sensitive to early replay buffer composition, especially in environments with delayed feedback or sparse rewards. Hybrid approaches that combine offline data with online fine-tuning aim to stabilize early learning by improving initial state-action coverage.

In this report, we investigate the effect of RLPD on SAC performance across multiple continuous control environments. All experiments in this work are con-

ducted entirely in simulation, and questions related to real-world deployment or sim-to-real transfer are considered out of scope.

2 Problem Formulation

2.1 State and Action Spaces

- **Hopper-v4:** 11-dimensional state space, 3-dimensional action space.
- **Ant-v5:** 111-dimensional state space, 8-dimensional action space.
- **AntMaze:** Ant robot navigation task with a high-dimensional continuous state space and an 8-dimensional continuous action space, requiring long-horizon planning.

2.2 Reward Function and Termination

- **Hopper-v4:** forward velocity minus control cost; episode terminates on falling or reaching the max step limit.
- **Ant-v5:** forward velocity reward plus survival bonus; episode ends on falling or reaching the max step limit.
- **AntMaze:** sparse reward of 1 upon reaching the goal and 0 otherwise; episodes terminate upon reaching the goal or the time limit.

2.3 Success Criteria

A run is considered successful if the agent achieves average episodic rewards comparable to baseline SAC performance.

3 System Design

This project adopts a purely simulation-based system design and does not involve physical robot hardware, real-world sensing, or sim-to-real transfer. Experiments were conducted using Gymnasium continuous control environments. Ant and Hopper long-

horizon experiments (1,000,000 steps) are included to study stability over extended training.

The SAC algorithm interacts with the environment using off-policy updates. For RLPD experiments, pre-collected datasets are loaded into the replay buffer prior to online training.

4 Methods

4.1 Algorithms

- SAC (Soft Actor-Critic) with automatic entropy tuning.
- RLPD: SAC initialized with pre-collected trajectories from partially trained agents.

4.2 Offline Data Collection

For RLPD experiments, offline datasets were collected from partially trained SAC agents at intermediate checkpoints. These datasets capture a range of behaviors from suboptimal to moderately successful policies, providing diverse exploration data for initialization. The replay buffer is pre-filled with these offline transitions before online training begins, allowing the agent to leverage prior experience immediately.

4.3 Training Details & Hyperparameters

Parameter	Description	Values
Replay Buffer Size	Max stored transitions	50k, 300k, 500k
Batch Size	Samples per update	64, 128, 256, 512
Learning Rate	Adam step size	3×10^{-4}
Discount Factor	Future reward weight	0.99
Training Steps	Environment steps	200k, 1M, 2M
Seeds	Random initialization	0–9

Table 1: Key hyperparameters across experiments.

4.4 Evaluation Metrics

- Average episodic reward across seeds.
- Standard deviation to quantify variability.
- Learning curves as a function of environment steps.

4.5 AntMaze Design Choice Experiments

AntMaze presents a significantly more challenging setting due to its sparse reward structure and long-horizon navigation requirements. To evaluate the effectiveness of RLPD-style techniques, we investigated

three design choices proposed in the original RLPD paper:

- **DC1:** Mixing offline and online data via a fixed offline ratio.
- **DC2:** Applying layer normalization within the critic network to reduce overestimation and training variance.
- **DC3:** Using entropy-regularized backup targets to encourage broader exploration.

We conducted four controlled experiments on AntMaze:

- Baseline using DC1 + DC2
- DC1 only
- DC1 + DC3
- DC1 + DC2 + DC3

All runs used identical hyperparameters and random seeds to isolate the effect of each design choice.

5 Experiments & Results

5.1 Quantitative Summary

Table 2 summarizes the final performance across environments and key configurations.

Environment	Configuration	Final Return
Hopper-v4	Buffer 50k	Stable convergence
Hopper-v4	Buffer 300k	Stable convergence
Hopper-v4	Buffer 500k	Stable convergence
Ant-v5	Multiple seeds	804.38 ± 379.62
AntMaze	DC1 only	Best early learning
AntMaze	DC1 + DC3	Most stable final
AntMaze	DC1 + DC2	Performance dip
AntMaze	DC1 + DC2 + DC3	Slowest learning
Walker2d	Random init	Slowest growth
Walker2d	50k prefill	Faster growth
Walker2d	100k prefill	Fastest growth

Table 2: Summary of final performance across environments.

Hopper-v4 shows stable convergence across buffer sizes. Ant-v5 exhibits high variance. AntMaze results highlight trade-offs between design choices. Walker2d demonstrates clear benefits from larger offline datasets.

5.2 Hopper-v4

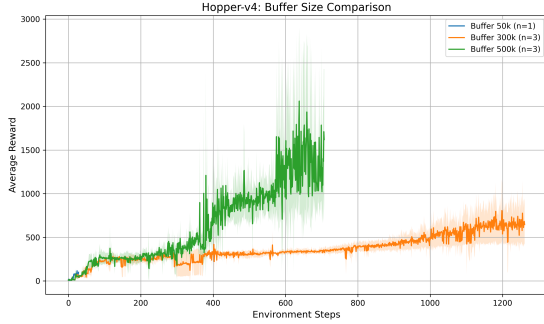


Figure 1: Hopper-v4 learning curves under varying buffer sizes (batch 256). Shaded areas represent one standard deviation across $n=3$ seeds.

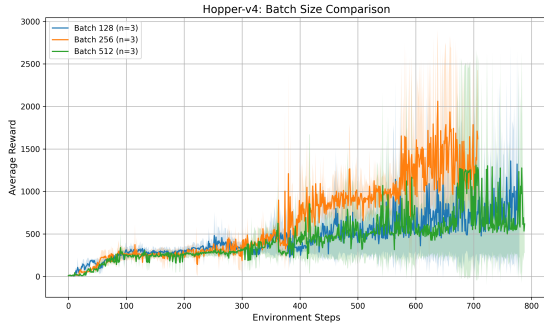


Figure 2: Hopper-v4 learning curves under varying batch sizes (buffer 500k). Mean $\pm$ std across seeds.

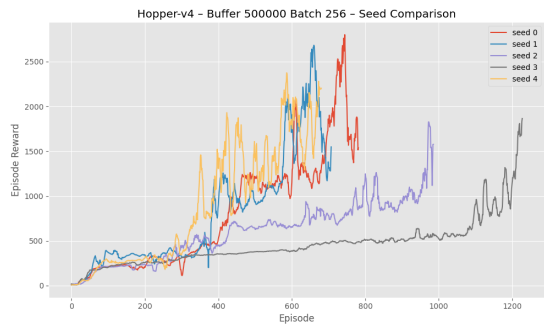


Figure 3: Seed-wise variability in Hopper-v4 with buffer 500k, batch 256. Highlights differences due to initialization and stochasticity.

Figures 1–3 show learning curves for Hopper-v4 under various configurations. Buffer size has minimal impact on final performance, with all configurations converging to similar reward levels. Batch size variations (Fig. 2) show that larger batches provide slightly smoother learning curves. Seed-wise variability (Fig. 3) demonstrates relatively consistent perfor-

mance across different random initializations, indicating stable learning dynamics in this environment.

Hopper-v4 demonstrates robust learning across all tested configurations. The consistency across different buffer sizes (Fig. 1) suggests that even modest amounts of offline data are sufficient for this environment. Larger batch sizes produce smoother learning curves, though all configurations achieve comparable final performance. The low seed sensitivity makes Hopper-v4 reliable for hyperparameter tuning and validation.

5.3 Ant-v5

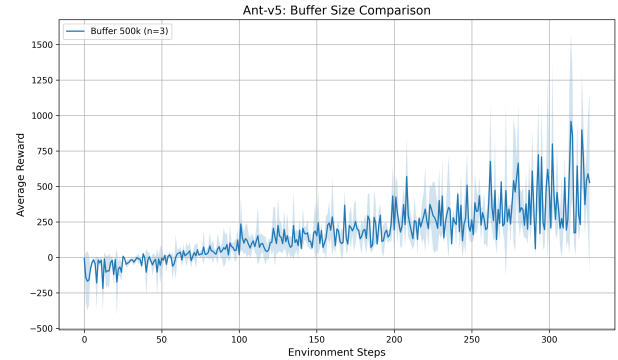


Figure 4: Ant-v5 learning curves across seeds (buffer 500k, batch 256). Shaded regions indicate one standard deviation. High-dimensional control results in more variability.

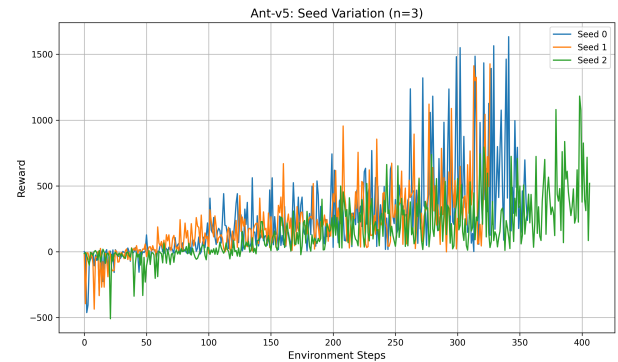


Figure 5: Seed-wise variability for Ant-v5. Significant divergence across seeds indicates sensitivity to initialization and early exploration. Mean return = 804.38, std = 379.62.

Ant-v5 results are shown in Figures 4–5. The higher-dimensional state and action spaces lead to increased variability compared to Hopper-v4. Figure 5 reveals substantial seed-dependent variance with a

mean return of 804.38 and standard deviation of 379.62, highlighting the challenge of achieving consistent performance in complex locomotion tasks. Despite this variability, RLPD provides more stable early-stage learning compared to purely online approaches.

The high variance in Ant-v5 stems from its 111-dimensional state space and 8-dimensional action space, which significantly expand the exploration challenge. Figure 5 shows distinct performance clusters across seeds, with some achieving returns above 1000 while others plateau below 500. This indicates high sensitivity to initial exploration trajectories and the difficulty of discovering coordinated locomotion gaits. RLPD’s benefit is most evident in early training, where offline data provides a more informative starting point than random exploration.

5.4 AntMaze

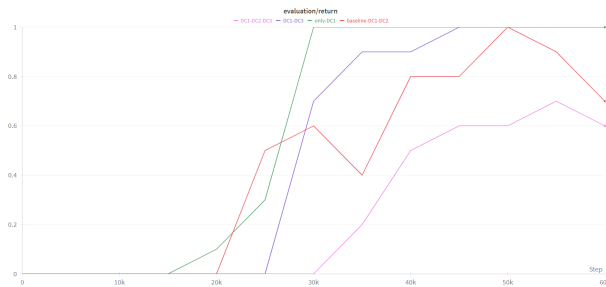


Figure 6: Evaluation return over training steps. Only-DC1 (green) reaches maximum return earliest with the cleanest convergence curve. Baseline-DC1-DC2 (red) experiences a significant dip around 35k steps. DC1-DC2-DC3 (pink) learns slowest among all configurations.

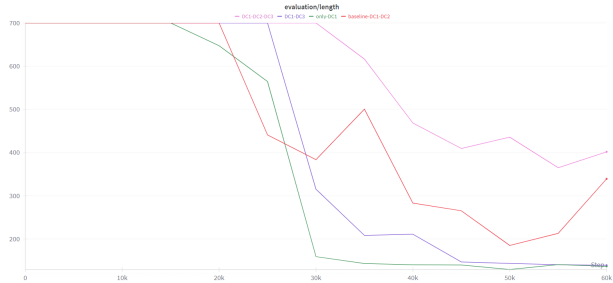


Figure 7: Evaluation episode length over training steps. Lower values indicate more efficient navigation to the goal. Only-DC1 (green) converges fastest to minimal length, while DC1-DC3 (blue) achieves the most stable final performance. Baseline-DC1-DC2 (red) shows a performance regression around 35k steps.

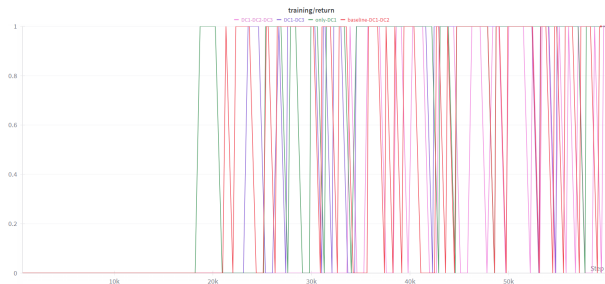


Figure 8: Training return during learning. Sparse successful episodes (return = 1.0) indicate the agent reached the goal during that training episode. All configurations show similar volatility throughout training, with frequent failures to reach the goal despite improving evaluation performance.

AntMaze results highlight trade-offs between sample efficiency and evaluation consistency in sparse-reward environments. Only-DC1 achieved the fastest early learning, reaching high evaluation returns around 28–30k steps with the cleanest return curve (Fig. 6). DC1-DC3 demonstrated the best final performance with the lowest and most stable episode lengths (Fig. 7). However, baseline-DC1-DC2 experienced a notable performance regression around 35k steps before recovering (Fig. 8).

Combining all three design choices (DC1-DC2-DC3) learned slowest and did not outperform simpler configurations by 60k steps, suggesting diminishing returns from stacking stabilization techniques. Training dynamics remained similarly volatile across all methods. These results indicate that while offline data accelerates learning, the effectiveness of stabi-

lization depends on the specific mechanism.

5.5 Walker2d

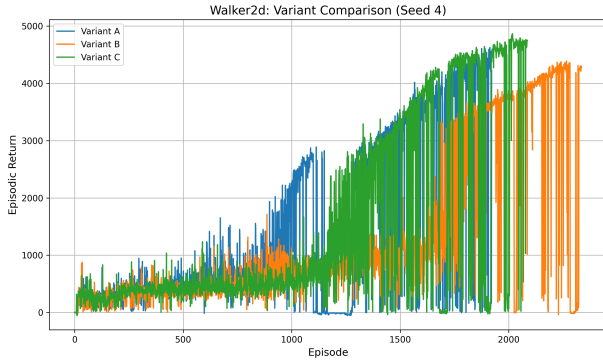


Figure 9: Walker2d learning curves comparing replay buffer initialization strategies (Seed 4).

Figure 9 compares three replay buffer initialization strategies. Variant A uses a randomly initialized replay buffer, Variant B is prefilled with 50k offline transitions, and Variant C is prefilled with 100k offline transitions. While none of the variants fully converged within the evaluated training horizon, Variants B and C exhibit substantially faster reward growth during the early and mid training stages, indicating improved sample efficiency relative to random initialization. The 100k prefill (Variant C) shows the most consistent improvement, suggesting that larger offline datasets provide a richer initialization signal and increase the likelihood of reaching strong performance earlier in training. Results are averaged over multiple random seeds; Walker2d results are reported for a single representative seed (Seed 4).

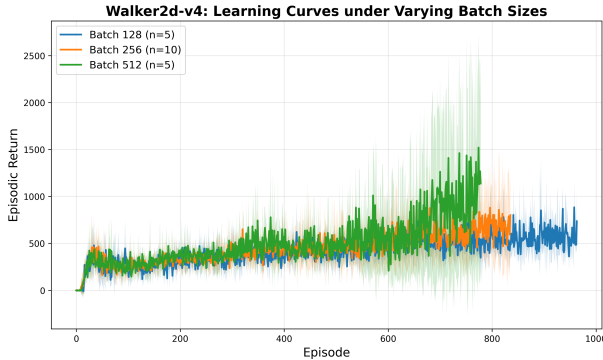


Figure 10: Walker2d batch size comparison across training steps.

Figure 10 shows training performance for batch

sizes of 128, 256, and 512 across multiple seeds. Although none of the configurations have clearly converged, larger batch sizes produce smoother learning dynamics and faster early reward growth. In particular, a batch size of 512 exhibits the lowest variance across seeds in the early training phase, suggesting improved stability during optimization.

6 Discussion

Hopper-v4 exhibited stable learning across random seeds, replay buffer sizes, and batch sizes. While RLPD accelerated early-stage learning, it did not significantly affect final performance, indicating that its primary benefit lies in improving early training efficiency rather than asymptotic returns.

Ant-v5 showed greater sensitivity to random seeds, producing noisier learning curves due to its higher-dimensional state and action spaces. In this environment, RLPD reduced early-stage instability and improved initial reward accumulation, suggesting that offline replay buffer initialization helps mitigate early exploration challenges.

Walker2d similarly benefited from offline replay buffer initialization, with both 50k and 100k prefill variants achieving faster reward growth than random initialization. The 100k prefill consistently outperformed the 50k variant, indicating that larger and more diverse offline datasets provide stronger early training signals and improved sample efficiency.

Across environments, replay buffer size and batch size were key factors influencing learning stability, underscoring the importance of careful data-related hyperparameter tuning. AntMaze introduced additional difficulty due to sparse rewards and long-horizon dependencies. In this setting, offline data alone enabled stable goal-reaching behavior, with the DC1 configuration achieving consistent evaluation performance. Entropy-regularized backups improved episode efficiency, whereas layer normalization introduced instability, and combining multiple stabilization mechanisms yielded diminishing returns.

7 Limitations and Failure Analysis

- Ant-v5 and Hopper long-horizon experiments are computationally intensive, limiting seed count.
- Offline data quality is dependent on checkpoint quality.
- AntMaze experiments were limited to 60,000 training steps due to computational constraints, compared to 300,000–1,000,000 steps typically used in RLPD benchmarks.

Several failure modes were observed during training. In AntMaze, agents occasionally exhibited performance regression after initial improvement, likely due to off-policy drift and value overestimation. Walker2d showed sensitivity to replay buffer initialization, with some variants failing to recover from early suboptimal behavior.

8 Future Work

Future work could increase the number of random seeds and extend training horizons to obtain more robust and statistically reliable conclusions, as well as to analyze long-term convergence behavior. In locomotion environments, longer training runs would enable examination of whether learned policies converge to stable gait patterns or exhibit performance degradation due to instability or value drift. Additionally, evaluating the effect of varying offline dataset quality would help clarify the sensitivity of replay buffer prefill strategies to data source and quality. Finally, comparing Soft Actor-Critic with other offline-capable reinforcement learning algorithms, such as TD3+BC and Implicit Q-Learning, would provide insight into whether the observed trends generalize beyond a single algorithm.

9 Conclusion

Using pre-collected datasets in SAC improves early-stage stability and reduces seed-dependent variance, especially in complex environments. RLPD-style approaches enhance sample efficiency in reinforcement learning and provide guidance for selecting hyperparameters in future experiments.

Our experiments across Hopper-v4, Ant-v5, Walker2d, and AntMaze demonstrate that the benefits of offline data are most pronounced during early training phases, where pre-filled replay buffers accelerate learning and reduce initial exploration challenges. The magnitude of improvement scales with both the quantity of offline data and the complexity of the environment.

References

- [1] Ball, P. J., Smith, L., Kostrikov, I., & Levine, S. (2023). *Efficient Online Reinforcement Learning with Offline Data*. <https://arxiv.org/abs/2302.02948>
- [2] RLPD GitHub Repository. <https://github.com/ikostrikov/rlpd>